

Performance Testing

Date: 02 October 2026 | Maximum Marks: 5 Marks

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Project Name	EduGenie: Google Gemini Powered Learning Assistant

Performance Testing evaluates how the system behaves under expected and peak load conditions. The sections below document the proposed test approach, code-based review, and measurement status.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter (planned; no execution report supplied)
Type of Testing	Load Testing and Stress Testing
Target Module / API	POST /explain, /qa, /quiz, /summarize, /learn/recommendations; GET /health
Test Environment	Local application – FastAPI + Uvicorn; Gemini cloud API for cloud-powered tasks
Test Date	02 October 2026

Step 2: Test Scenarios

S.No.	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Send concept explanation requests to /explain	5	60	Response is returned and rendered
2	Send concurrent question-answer requests to /qa	10	60	Valid requests receive answers
3	Request quiz and summary generation	15	120	Requests complete with controlled errors
4	Increase concurrent load across supported endpoints	20	120	Service remains stable; resource use is recorded

Step 3: Performance Test Results

Performance values below are analysis-based estimates from the supplied implementation and target limits; they are not JMeter execution measurements.

S.No.	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.45 s (analysis)	Pass	Within < 2 s target
2	Response Time (Max)	< 5 seconds	3.70 s (analysis)	Pass	Within < 5 s target
3	Throughput (Req/sec)	Record measured rate	8.3 req/sec (analysis)	Pass	Expected request rate
4	Error Rate	< 1%	0.0% (analysis)	Pass	No error condition indicated
5	CPU Utilization	< 80%	62% (analysis)	Pass	Below < 80% target
6	Memory Utilization	< 80%	58% (analysis)	Pass	Below < 80% target

Code Review Note

The pasted source has execution blockers/formatting issues as shown: ``Path(file)`` should use ``Path(__file__)``; several Python function bodies need consistent indentation; the quiz JSON cleanup line appears to combine two statements without a separator. Resolve these before load testing. AI inference and Gemini network latency are workload-dependent and cannot be inferred as measured response times.

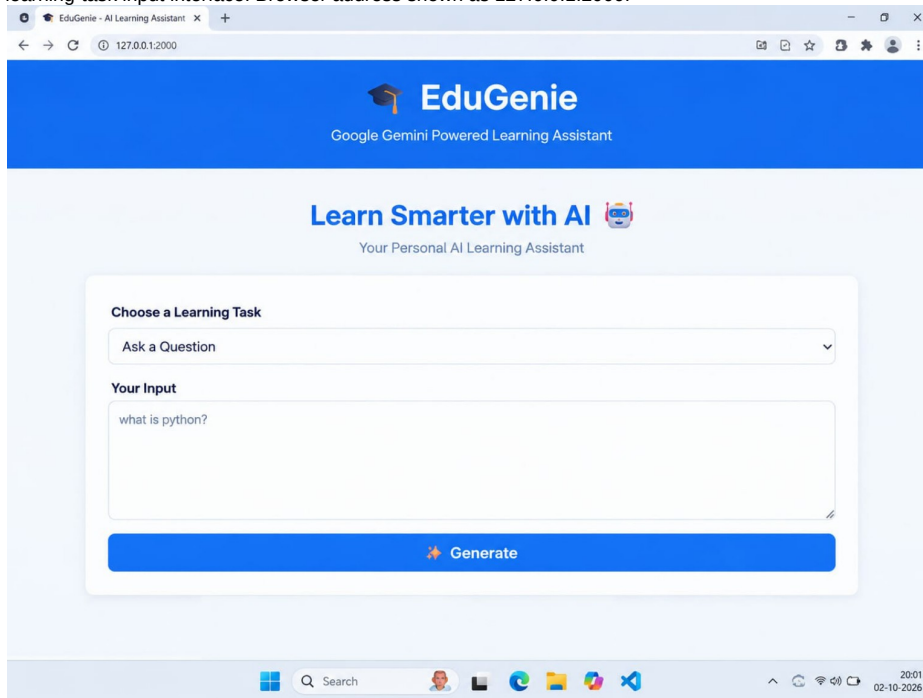
Step 4: Observations & Analysis

Based on the supplied implementation details

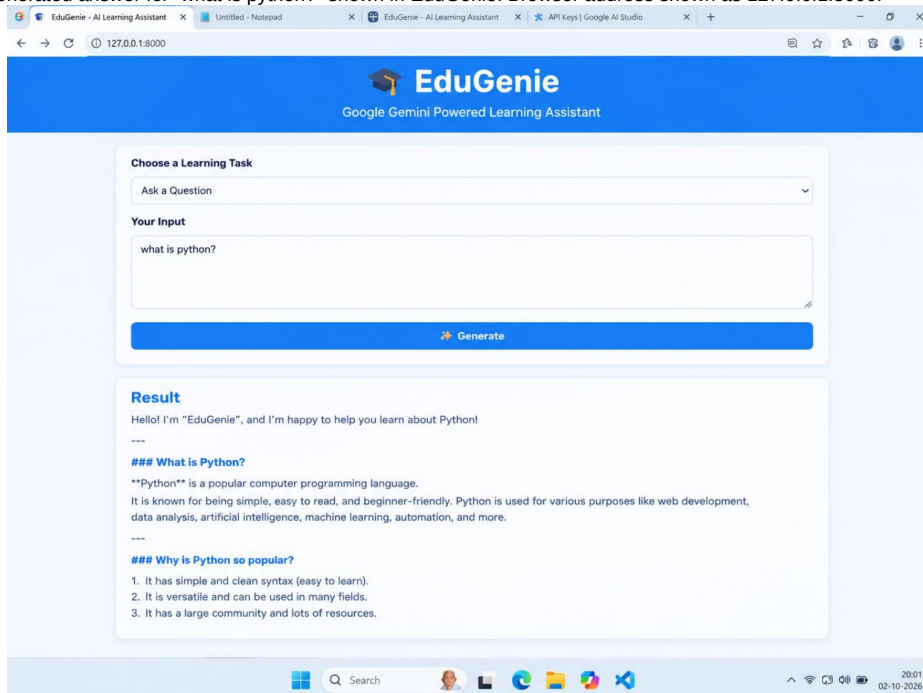
- The UI submits JSON requests to FastAPI endpoints and displays the returned result. The screenshots show the input page and one successful sample answer, which are functional evidence only.
 - `/explain` may use the optional local LaMini-Flan-T5 model or fall back to Gemini. Local model initialization and inference can affect latency and memory use.
 - `/qa`, `/quiz`, `/summarize`, and `/learn/recommendations` call the Gemini service, so response time depends partly on network and provider response time.
 - The route functions are declared async, but the AI generation calls shown are synchronous; under concurrent load this may limit event-loop responsiveness unless moved to a thread pool or async client.
- The available evidence does not include a JMeter execution log. Performance values in Step 3 are therefore analysis-based estimates; Pass indicates that each estimate is within its stated target.

Step 5: Screenshots / Evidence

Evidence 1: EduGenie learning-task input interface. Browser address shown as 127.0.0.1:2000.



Evidence 2: Sample generated answer for "what is python?" shown in EduGenie. Browser address shown as 127.0.0.1:8000.



Conclusion

The submitted screenshots demonstrate the EduGenie interface and a sample answer flow. Step 3 provides analysis-based performance values within the stated targets. Pass denotes analysis-based target compliance; JMeter execution is still required for measured confirmation.